INCH-POUND

MIL-DTL-495F
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SUPERSEDING
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30 September 1988

DETAIL SPECIFICATION

FINISH, CHEMICAL, BLACK, FOR COPPER ALLOYS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers a black chemical finish for copper alloys.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A-50153 - Detergent, Laundry, Nonionic, Liquid

(Copies of this document are available online at https://quicksearch.dla.mil.)

Comments, suggestions, or questions on this document should be addressed to U.S. Army Ground Vehicle Systems Center, ATTN: Standardization: FCDD-GVS-SAT, MS #268, 6501 E. 11 Mile Road, Detroit Arsenal, MI 48397-5000 or emailed to usarmy.detroit.devcom-gvsc.mbx.standardization@army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ Z1.4	-	Sampling for Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at www.asq.org.)

ASTM INTERNATIONAL

ASTM D523	-	Standard Test Method for Specular Gloss
ASTM D822/D822M	-	Standard Practice for Filtered Open-Flame Carbon-Arc Exposures of Paint and Related Coatings
ASTM G152	-	Standard Practice for Operating Open Flame Carbon Arc Light Apparatus for Exposure of Nonmetallic Materials

(Copies of these documents are available online at www.astm.org.)

INTERNATIONAL AEROSPACE ENVIRONMENTAL GROUP (IAEG)

AD-DSL	-	Aerospace and Defense Declarable Substance List

(Copies of this document are available online at www.iaeg.com/chemicalrpt/addsl.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, superseded applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Manufacturers are given latitude in the selection of raw materials provided the finish meets the requirements of this document.

3.1.1 Toxic chemicals, hazardous substances, and ozone-depleting chemicals. The final product shall contain no chemicals labelled “R1” or “R2” on the AD-DSL list. The content of the AD-DSL and its application does not relieve the plating applicator from obligation to comply with all existing relevant regional and national regulations.

3.2 Preparation of surface prior to application of black finish. Surfaces to be finished shall be thoroughly cleaned to remove all grease, oil, scale, and foreign matter. If desired, surfaces may be given additional preparatory treatments to assist in meeting the requirements of this document.

3.3 Black chemical finish. The black chemical finish shall be produced by chemical or electrochemical treatment of the copper-alloy surface. Hexavalent chromium shall not be used.

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3.3.1 Stress-corrosion prevention. When specified (see 6.2), copper alloys with high residual stresses shall be finished by a process excluding ammonia.

3.4 Rinsing. Black finish surfaces shall be thoroughly rinsed free of treatment chemicals and shall be dried. Care shall be exercised to effect thorough rinsing and drying in crevices and recesses of the articles being finished. Chromic acid shall not be used in sealing or rinsing.

3.4.1 Paint base (not for Army ground vehicle use). Where the black chemical finish is to be used as a base of lacquer or other paint coating (see 6.2), the rinsing shall be followed by a 30-second dip in a 1/2 ounce of chromic acid[/1] per gallon of water solution. The acid dip shall be followed by rinsing with water and drying.

[/1] Chromic acid is an R1, R2, and D1 hazardous substance in the Aerospace and Defense Declarable Substance List. It shall not be used on parts for Army Ground Vehicles.

3.5 Performance.

3.5.1 Resistance to hot soap solution. The finish shall show no more than a slight change in color and appearance and shall not loosen, be removed, or flaked when tested as specified in 4.3.2.2.

3.5.2 Resistance to accelerated weathering. The finish shall show no more than a slight change in color and appearance and shall not have any finish loosening, removal, or flaking when tested as specified in 4.3.2.3.

3.5.3 Flexibility. Where the article is to be clinched or bent subsequent to the finishing operation (see 6.2), the finish shall withstand clinching and bending stresses without cracking, except at those points where cracking in the metal itself is permissible when tested as specified in 4.3.2.4.

3.6 Color and gloss. The finish shall be black and shall completely hide the color of the underlying material. Gloss shall be 25 maximum when tested as specified in 4.3.2.1. The gloss requirement shall not apply when the chemical finish is to be coated with oil, wax, lacquer, or other films (see 6.2).

3.7 Workmanship. The finish shall be uniform and, unless otherwise specified, small variations in appearance will be permitted on surfaces not significantly exposed to view. The finish shall be continuous, clean, and free from flaking, breaks, or scratches.

4. VERIFICATION

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use their own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of section 3. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirement in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptable of defective material.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with ASQ Z1.4.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.2.2 In-process inspection. When stress-corrosion prevention is specified (see 3.3.1), the finishing operation shall be examined for conformance with the requirements of 3.3.1. Whenever nonconformance is noted, correction shall be made to the operation and all affected items shall be rejected.

4.2.3 End item visual examination. The end item shall be examined for conformance to the workmanship requirements of 3.7. The lot size shall be expressed in units of black chemical finished items. The sample unit shall be one finished item. The inspection level shall be I and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 4.0.

4.2.4 End item testing. End item testing shall be tailored to meet the service requirements of the end item. Test methods are listed in 4.3.2. The specific test methods required for an item shall be specified on the drawings or in the contract. When testing is specified, (see 6.2), the inspection level shall be S-1, and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.3 Methods of inspection.

4.3.1 Test specimens. When the size, shape or cost of the finished item is such that tests cannot be performed on the item, sample test specimens may be used for tests 4.3.2.1 through 4.3.2.4. Such test specimens shall be cut from scrap items or made from the same copper alloy and finished by the same process as the end item. The test specimens shall have a minimum external surface of 4 square inches. Specimens shall be randomly distributed throughout the processed lot and finished concurrently with the items. Test specimens shall be used only once.

4.3.2 Procedures.

4.3.2.1 Gloss. Gloss shall be measured in accordance with ASTM D523. On articles which are too small for proper measurement by glossmeter and on articles with curved surfaces, the gloss shall be estimated visually against a black panel having a gloss of 25. In this visual determination, the high-light effect due to sharp curvatures shall be ignored. Any nonconformance to the requirements in 3.6 shall constitute a test failure.

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4.3.2.2 Resistance to hot soap solution. The surface to be tested shall be immersed in a 5 percent soap solution at 195 °F for 2 hours, then rinsed and examined. The soap used shall conform to A-A-50153. Where the black finish is intended as a base for other coatings, the test shall be conducted on uncoated black finished surfaces. Any nonconformance to the requirements in 3.5.1 shall constitute a test failure.

4.3.2.3 Resistance to accelerated weathering. The surface to be tested shall be exposed for 200 hours in accordance with ASTM D822/D822M and ASTM G152. The test sample shall be rinsed using the water supply, spray nozzles, and pressure as prescribed in the standard test method. The sample shall be lightly washed under the spray using a thoroughly degreased lambs wool or absorbent cotton pad. The excess water shall be wiped off with a clean lint free cloth, and the sample dried for at least two hours and examined for conformance to the requirements in 3.5.2. Any nonconformance shall constitute a test failure. Where the black finish is intended as a base for other coatings, the test shall be conducted on uncoated black finished surfaces.

4.3.2.4 Flexibility. The sample item shall be flexibly tested by bending as in the assembled condition. When a test specimen is used, it shall be bent 180 degrees over a 1/2 inch diameter rod. Any nonconformance to the requirements in 3.5.3 shall constitute a test failure.

5. PACKAGING

This section is not applicable to this specification.

6. NOTES

6.1 Intended use. The black chemical finish is intended to be used to impart a black color to and for gloss reduction purposes on copper-alloy surfaces other than food service and water supply items. It is also used as a base for subsequent coatings such as lacquer, varnish, oil, and wax.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this document.
b. When ammonia is to be excluded from finished process (see 3.3.1).
c. When the finish is to be used as a base for another coating (see 3.4.1).
d. When the item is to be clinched or bent after finishing (see 3.5.3).
e. When the finished item is to be coated with oil, wax, lacquer, or other film (see 3.6).
f. When end item testing is required (see 4.2.4).

6.3 Subject term (key word) listing.

Coating base
Chromic acid

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6.4 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

CONCLUDING MATERIAL

Custodians:
Army – AT
Air Force – 20

Preparing Activity:
Army – AT

(Project MFFP-2024-003)

Review Activities:
Army – AR, CR4, EA

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